

AI & ALGORITHMS USED

Smart Public Grievance Escalation System

1. RULE-BASED AI — Priority Assignment

What is it?

A **Rule-Based Expert System** — one of the oldest and most reliable forms of AI. It mimics how a human expert makes decisions using a set of IF-THEN rules.

Where is it implemented?

File: `backend-node/server.js` **Function:** `assignPriority(description, category)` — Line 160

How it works:

```
STEP 1: User submits complaint
      ↓
STEP 2: System reads:
      - Category (Water Supply, Health, Roads...)
      - Description text (emergency, broken, urgent...)
      ↓
STEP 3: Apply Rules:

IF category contains "water" OR "electricity" OR "health" OR "safety"
  → Priority = HIGH

ELSE IF description contains "urgent" OR "emergency" OR "critical" OR "danger"
  → Priority = HIGH

ELSE IF category contains "roads" OR "sanitation" OR "transportation"
  → Priority = MEDIUM

ELSE IF description contains "broken" OR "damaged" OR "not working"
  → Priority = MEDIUM

ELSE
  → Priority = LOW
      ↓
STEP 4: Priority assigned automatically (High / Medium / Low)
```

Actual Code:

```
function assignPriority(description, category) {
  const descLower = description.toLowerCase();
```

```
const catLower = category.toLowerCase();

// HIGH priority keywords
const highKeywords = ['urgent', 'emergency', 'critical', 'danger',
                      'life', 'death', 'severe', 'immediate',
                      'accident', 'fire', 'flood', 'leak'];

// HIGH priority categories
const highCategories = ['health', 'safety', 'water', 'electricity'];

if (highKeywords.some(k => descLower.includes(k)) ||
    highCategories.some(c => catLower.includes(c))) {
  return 'High';
}

// MEDIUM priority categories
const mediumCategories = ['roads', 'sanitation', 'transportation'];
if (mediumCategories.some(c => catLower.includes(c))) {
  return 'Medium';
}

// MEDIUM priority keywords
const mediumKeywords = ['broken', 'damaged', 'not working',
                        'repair', 'fix', 'bad', 'poor', 'faulty'];
if (mediumKeywords.some(k => descLower.includes(k))) {
  return 'Medium';
}

return 'Low';
}
```

Priority Rules Table:

Priority	Categories	Keywords
HIGH	Water Supply, Electricity, Health, Safety	urgent, emergency, critical, danger, fire, flood, leak, accident
MEDIUM	Roads, Sanitation, Transportation	broken, damaged, not working, repair, fix, bad, poor, faulty
LOW	Education, Other	(everything else)

Why Rule-Based AI?

- Simple and transparent — easy to explain
- No training data needed
- 100% predictable results
- Fast execution (milliseconds)
- Easy to update rules
- Used in real government systems worldwide

2. SLA-BASED ESCALATION ALGORITHM

What is it?

A **Time-Based Automated Decision Algorithm** that monitors complaint deadlines and automatically escalates overdue complaints without human intervention.

Where is it implemented?

File: `backend-node/server.js` **Function:** `checkAndEscalateComplaints()` — Line 418 **Trigger:** `setInterval()` — runs every 5 minutes

How it works:

```
EVERY 5 MINUTES:
    ↓
STEP 1: Scan all complaints in database
    ↓
STEP 2: Check condition:
    IF complaint.status NOT IN (Resolved, Closed)
    AND complaint.sla_deadline < current_time
    AND complaint.escalation_level < 2
    ↓
STEP 3: Escalate:
    - Change status → "Escalated"
    - Increment escalation_level (0 → 1 → 2)
    - Log in escalation_log table
    - Send notification to user
    ↓
STEP 4: Repeat after 5 minutes
```

SLA Deadline Calculation:

File: `backend-node/server.js` **Function:** `calculateSLADeadline(priority)` — Line 195

```
HIGH Priority    → Deadline = Now + 24 hours
MEDIUM Priority → Deadline = Now + 48 hours
LOW Priority     → Deadline = Now + 72 hours
```

Actual Code:

```
function calculateSLADeadline(priority) {
    const now = new Date();
    let hours = 72; // Default Low priority

    if (priority === 'High')    hours = 24;
    if (priority === 'Medium') hours = 48;
```

```
    return new Date(now.getTime() + hours * 60 * 60 * 1000);  
  }  
  
  // Runs every 5 minutes  
  setInterval(checkAndEscalateComplaints, 5 * 60 * 1000);
```

Escalation Flow:

```
Complaint Submitted  
  ↓  
SLA Deadline Set (24h / 48h / 72h)  
  ↓  
Officer Works on Complaint  
  ↓  
[If NOT resolved before deadline]  
  ↓  
Auto-Escalation Triggered  
  ↓  
Status → "Escalated"  
  ↓  
User Notified  
  ↓  
[If STILL not resolved]  
  ↓  
Level 2 Escalation (max)
```

3. INTELLIGENT OFFICER ASSIGNMENT ALGORITHM

What is it?

A **Category-Based Matching Algorithm** that automatically assigns complaints to the most suitable department officer.

Where is it implemented?

File: `backend-node/server.js` **Function:** `autoAssignOfficer(category)` — Line 209

How it works:

```
STEP 1: User submits complaint with category "Water Supply"  
  ↓  
STEP 2: System queries database:  
  "Find officer WHERE department = 'Water Supply'"  
  ↓  
STEP 3a: If matching officer found  
  → Assign complaint to that officer  
  ↓
```

```
STEP 3b: If NO matching officer found
→ Assign to any available officer (fallback)
↓
STEP 4: Officer receives notification
```

Department Matching:

Complaint Category	Assigned Officer
Water Supply	water.officer@gov.in
Electricity	electricity.officer@gov.in
Roads	roads.officer@gov.in
Health	health.officer@gov.in
Sanitation	sanitation.officer@gov.in
Safety	safety.officer@gov.in

Actual Code:

```
async function autoAssignOfficer(category) {
  // Find officer matching the department
  const [officers] = await pool.query(
    'SELECT id FROM users WHERE role = "officer" AND department = ?
    ORDER BY RAND() LIMIT 1',
    [category]
  );

  if (officers.length > 0) {
    return officers[0].id; // Return matched officer
  }

  // Fallback: assign to any available officer
  const [anyOfficer] = await pool.query(
    'SELECT id FROM users WHERE role = "officer" ORDER BY RAND() LIMIT 1'
  );

  return anyOfficer.length > 0 ? anyOfficer[0].id : null;
}
```

4. NOTIFICATION INTELLIGENCE SYSTEM

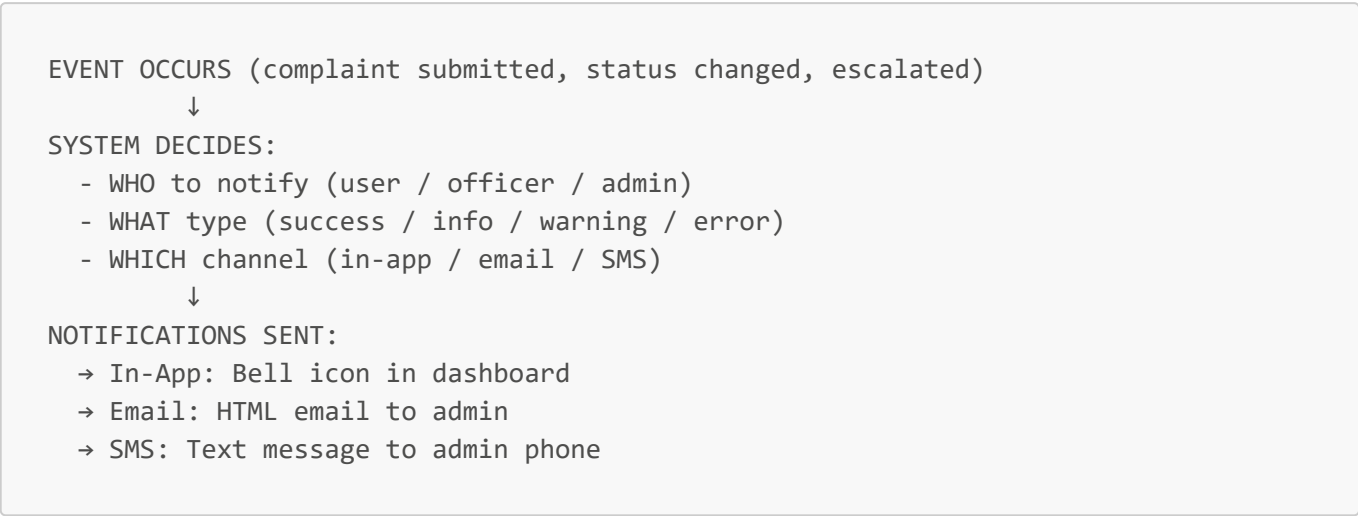
What is it?

A **Multi-Channel Event-Driven Notification System** that automatically sends the right notification to the right person at the right time.

Where is it implemented?

File: backend-node/server.js **Function:** createNotification(), notifyAdminAboutComplaint()

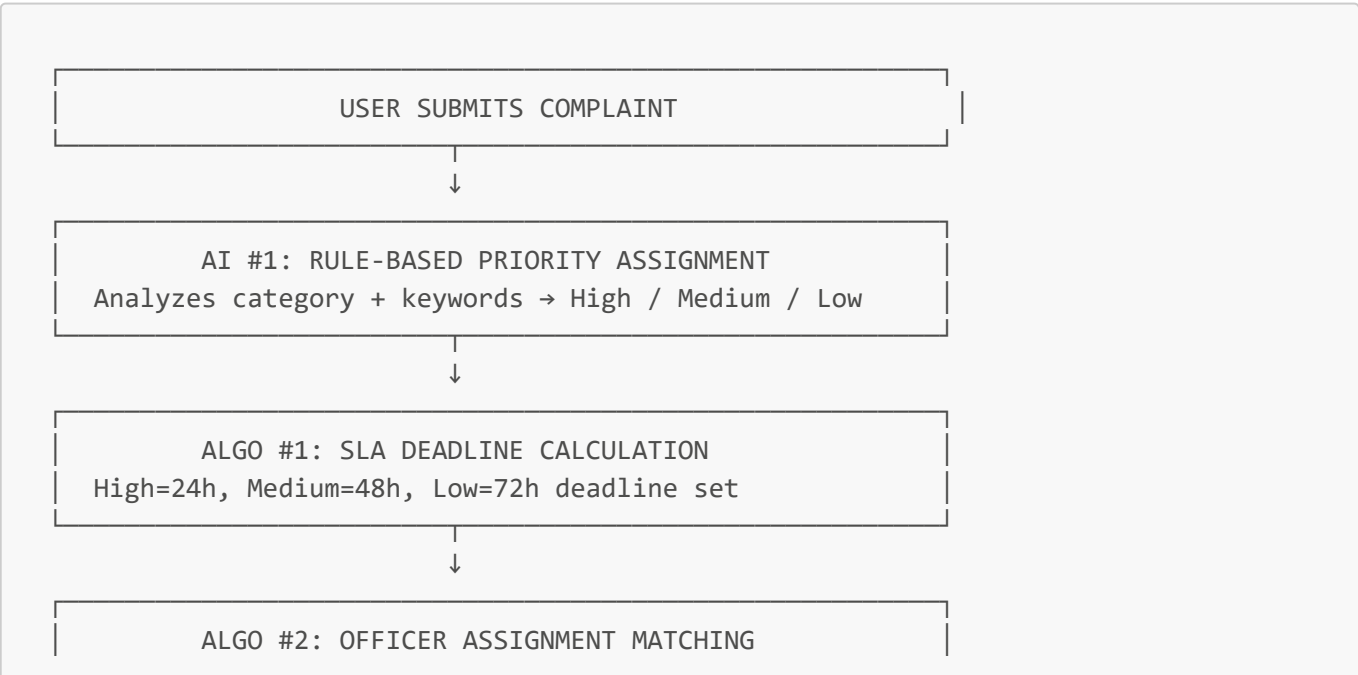
How it works:

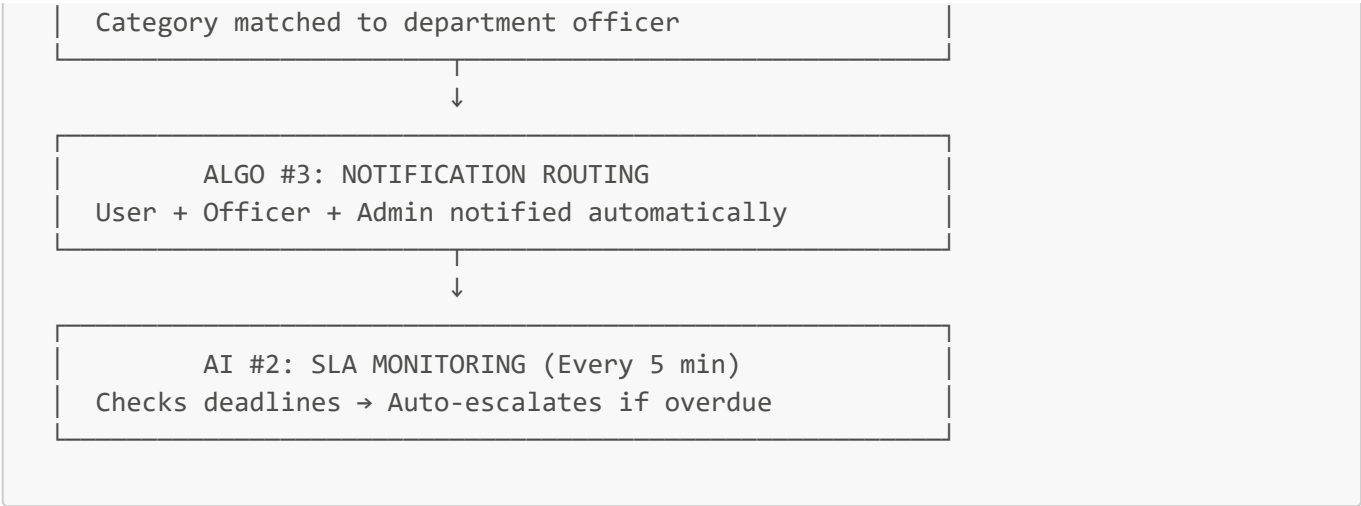


Notification Decision Table:

Event	Who Gets Notified	Type	Channel
Complaint submitted	User	Success	In-App
Complaint assigned	Officer	Info	In-App
Status updated	User	Info/Success	In-App
SLA escalated	User	Warning	In-App
New complaint	Admin	Warning	In-App + Email + SMS

5. COMPLETE AI SYSTEM FLOW





6. SUMMARY TABLE

#	AI / Algorithm	Type	File	Function	Purpose
1	Priority Assignment	Rule-Based AI	server.js	assignPriority()	Auto-assign High/Medium/Low
2	SLA Calculation	Mathematical Algorithm	server.js	calculateSLADeadline()	Set response deadlines
3	Auto-Escalation	Time-Based Algorithm	server.js	checkAndEscalateComplaints()	Escalate overdue complaints
4	Officer Assignment	Matching Algorithm	server.js	autoAssignOfficer()	Assign to right department
5	Notification Routing	Event-Driven Algorithm	server.js	notifyAdminAboutComplaint()	Multi-channel notifications

7. AI TYPE EXPLANATION

Rule-Based AI (Expert System)

- **Definition:** AI that uses predefined IF-THEN rules to make decisions
- **Used for:** Priority assignment
- **Advantage:** Transparent, fast, reliable, no training needed
- **Real-world use:** Medical diagnosis systems, loan approval systems, government portals

Algorithm (Non-AI but Intelligent)

- **Definition:** Step-by-step procedure to solve a problem
- **Used for:** SLA calculation, officer matching, escalation
- **Advantage:** Deterministic, predictable, efficient

8. FOR YOUR REVIEW PRESENTATION

Key Points to Mention:

1. "We implemented a **Rule-Based AI Expert System** for priority assignment"
2. "The AI analyzes both the **category** and **description keywords** to determine priority"
3. "We use a **Time-Based Escalation Algorithm** that runs every 5 minutes"
4. "An **Intelligent Matching Algorithm** automatically assigns complaints to the right officer"
5. "The system uses **Event-Driven Notification Routing** for multi-channel alerts"

Technical Terms to Use:

- Rule-Based Expert System
- Natural Language Keyword Analysis
- SLA (Service Level Agreement) Monitoring
- Automated Escalation Pipeline
- Intelligent Department Matching
- Event-Driven Architecture

File: AI_AND_ALGORITHMS.md Project: Smart Public Grievance Escalation System Last Updated: March 2026